

Assignment 1

Total Marks = 50

Answer the following questions directly on this paper and hand in your answers.

No electronic submissions are accepted. Your pages should be stapled together, with identification *on every page* (student number at least). A neat and professional appearance is expected and marks will be deducted otherwise.

General notes.

- A word-addressable system (or portion of) implies that the smallest location which can be addressed and whose content can be transferred is exactly one word, whatever size the word is.
- A byte-addressable system (or portion of) implies that the smallest location which can be addressed and whose content can be transferred is exactly one byte.
- Reminder: $1K=1,024=2^{10}$, $1M=1,024K=2^{20}$ and $1G=1,024M=2^{30}$
- Most answers should be computed and given as power of 2, or as power of 2 with a factor.

Question 1. [4] Consider a hypothetical 32-bit microprocessor having 32-bit instructions composed of two fields: the first byte contains the opcode and the remainder the immediate operand or an operand address.

(a) What is the maximum directly addressable space capacity (in bytes)? 2^{24}

(b) What is the maximum number of instructions which can be supported? 2^8

(c) What is the smallest size needed in bits for the program counter and the instruction register?

PC: 24 bits IR: 32 bits

Question 2. [7] The Motorola 68000 microprocessor is byte addressable with a 24-bit address bus and a 16-bit data bus, and one word contains 2 bytes. Your manager wants to purchase RAM chips each containing 1M words, where each word is 2 bytes, for a total of 2^{20} words (equivalent to 2^{21} bytes).

(a)[1] What is the maximum number of these RAM chips you could place in your system and be able to address?

$2^{24}/2^{21} = 2^3 = 8$ Chips

(b)[3] Having decided how much RAM could be bought, your manager realizes that the whole address space has been used up for it, leaving no room for peripherals to make the system useful after all. The manager tells you to buy half as much RAM as what was calculated above and use the rest of the address space to install peripherals. They would like to have at least 32 peripherals each of which is byte addressable and requires an address space of 64KB. Your manager asks you to check that it is all feasible..

Is it? **YES** NO (circle one)

Show a complete logical explanation of the reasoning for your answer, as you might do at work

8 chips * 2 mega bytes each / 2 = 8 mega bytes (2^{23}) available free.

32 peripherals (2^5) * $2^{16} = 2^{21}$

$2^{21} < 2^{23}$

- (c) If the answer above stated that 32 peripherals is too many, state how many you could handle in your system, without reducing further the RAM.
- (d) If the answer above stated that 32 peripherals fit exactly into the address space, then you are finished.
- (e) If the answer above stated that you could fit 32 peripherals and you had spare address space, state how much more RAM you could have in your system (that is, how many more chips you could buy).

Choose one of the three cases (c), (d) or (e) and give an answer with explanation. [3]

$$2^{23} - 2^{21} = 8\text{MB} - 2\text{MB} = 6\text{MB} = 2^{22} + 2^{21}$$

$$2^{22}/2^{21} = 2^1 = 2 \text{ Chips}, \quad 2^{21}/2^{21} = 2^0 = 1 \text{ Chip}$$

$$2 + 1 = 3 \text{ chips}$$

Question 3. [2] What are the lowest and highest addresses in a 2^{20} byte memory which a four-byte word is the smallest addressable unit?

$$2^{20}/2^2 = 2^{18}$$

$$0x00000, \text{ and } 0x3FFFF$$

Question 4. [5] In Figure 1, a sample port I/O bus architecture is shown, with the size of the address buses and data buses labelled. An 8-bit data bus denotes here a byte-addressable read/write cycle, while a 16-bit data bus denotes here a word-addressable read/write cycle, where a word here is 2 bytes in size.

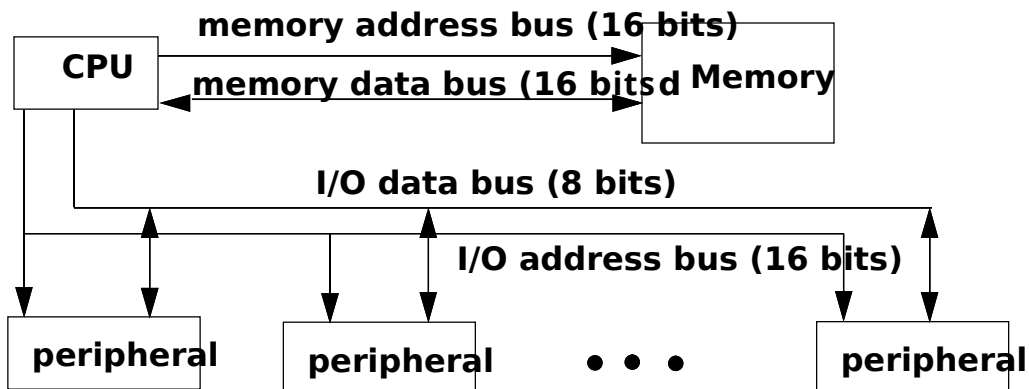


Figure 1: Port I/O sample bus structure

(a) How many addressable <i>locations</i> are available for memory?	2^{16}
(b) What is the size of each <i>location</i> ?	2 bytes
(c) What is the maximum memory size <i>in bytes</i> ?	$2^{16} \times 2^1 = 2^{17} \text{ bytes}$

(d)How large is the address space for I/O peripherals <i>in bytes</i> ?	2 ¹⁶ bytes
(e)If every peripheral must be allocated an address space of 2 ¹² bytes, what is the largest number of peripherals that can be connected?	2 ¹⁶ /2 ¹² = 2 ⁴

Question 5. [6] Sometimes software optimization can dramatically improve the performance of a computer system. Assume that a CPU can perform a multiplication operation in *10 ns*, and an addition or subtraction operation in *1 ns*. Assume that the operands (*x* in the question below) are already loaded into registers.

(a) How long will it take for the CPU to calculate the result of $x^2 + 2x$? 2*mult + 1*add = 21ns

(b) Could you optimize the equation so that it will take less time? Show all your steps and comment your reasoning in order to get full marks.

$x(x+2) \rightarrow 1\text{mult} + 1\text{add} = 11\text{ns}$

Question 6. [6] (a) [4] Give a short sequence of machine instructions for the task “*Add the contents of memory location A to those of memory location B, and place the answer in location C*”. You have available the following instructions:

Load Ri, LOC
Store Ri, LOC

which are the **only** instructions to transfer data between the memory and the general purpose registers. After operands have been loaded from memory into processor registers, arithmetic computation can be performed, using the instructions:

ADD Ri, Rj, Rk
SUB Ri, Rj, Rk

where i, j, k need not be distinct. Do not change the content of either location A or B.

Load R1, [A]
Load R2, [B]
ADD R3, R1, R2
Store R3, [C]

(b)[4] Suppose now that you have Move and Add instructions available with the formats:

MoveLOC1, LOC2

ADD LOC1, LOC2

These instructions move or add a copy of the operand at the second location to the first location, overwriting the original operand at the first location. Either or both of the operands can be in the memory or the general purpose registers. Is it possible to use fewer instructions of these types to accomplish the task of part (a)? If yes, give the sequence.

Move [C], [A]
ADD [C], [B]

Question 7. [8] Consider a system that has a byte-addressable memory organized in 32-bit words according to the big-endian scheme. A program reads ASCII characters entered at a keyboard and stores them in successive byte locations, starting at location 1000.

(a)[4] Show the contents of the two memory words at locations 1000 and 1004 after the name “Johnson” has been entered. Label each byte precisely with its content and its address.

character	J	o	h	n	s	o	n	
content in hex	4A	6F	68	6E	73	6F	6E	00
byte address	1000	1001	1002	1003	1004	1005	1006	1007

▲ WORD @ 1000 ▲ WORD @ 1004 ▲

(b)[4] Repeat the question above for the little-endian scheme.

character	n	h	o	J		n	o	s
content in hex	6E	68	6F	4A	00	6E	6F	73
byte address	1000	1001	1002	1003	1004	1005	1006	1007

▲ WORD @ 1000 ▲ WORD @ 1004 ▲

Question 8. [12] The Motorola 68000 microprocessor is byte addressable with a 24-bit address bus and a 16-bit data bus, and one word contains 2 bytes. There are both peripherals and memory units to be connected to build a system and it is expected that the whole address space will be used. The diagram below may help you visualize the required answers to the questions. State the answers in the appropriate boxes. Leave the values in the answers as powers of 2 or simple approximations to them or simple factors - there is no need to give the actual number.

(a) [1] What is the total number of addressable locations for this system?

$$(a) \quad 2^{24}$$

(b) [1] 1/4 of the addressable space is to be used for the peripherals requirements. What is the total number of available addresses for peripherals?

$$(b) \quad 2^{24}/2^2 = 2^{22}$$

The addressable space for peripherals is further subdivided as shown below.

(f) [1] 1/16 of the addresses are to be used for input components. How many?

$$(f) \quad 2^{22}/2^4 = 2^{18}$$

(g) [1] 1/16 of the addresses are to be used for output components. How many?

$$(g) \quad 2^{18}$$

(h) [1] 14/16 of the addresses are to be used for disk addressing. How many?

$$(h) \quad 2^{18} * (2^3 + 2^2 + 2^1) = 2^{21} + 2^{20} + 2^{19}$$

(c) [1] 3/4 of the addressable space is to be used for the memory requirements of RAM and ROM. What is the total number of addresses available for memory?

$$(c) \quad 2^{22} * (2^1 + 2^0) = 2^{23} + 2^{22}$$

(e) [1] How many addresses are left for RAM?

$$(e) \quad 16\text{MB} - 8\text{MB} - 4\text{MB} - 16\text{KB} = 2^{22} - 2^{14}$$

(d) [1] The ROM has already been bought and its size is 16KB. How many addresses does it need?

$$(d) \quad 2^{14}$$

(i) [1] The chosen disk drives each require 1MB of address space. How many can one fit into the system?

$$(i) \quad 2^{21} + 2^{20} + 2^{19} = 3.5 = 3$$

Following up from (e):

(e) How many addresses are left for RAM?

(e)

(j) [1] We want to buy a giant screen with a maximum resolution of 1440x900 pixels, where each pixel needs 1 byte of storage. It is normal to buy a RAM buffer to support such a display. How big must this be? Approximate to a power of 2, in bytes, which covers the display requirements.

(j)

(k) [1] How many addresses are left now for RAM?

(k)

(m)

(m) [1] The cheapest RAM chips contain 1M words. How many chips can you buy to fit into this system?